

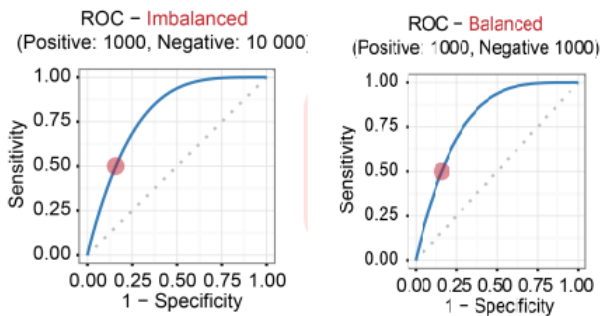
Machine Learning (CSE543/ECE543)

Practice Questions

1. Given a classification problem with input features as shown in the graph. Can we use logistic regression to classify this dataset? Explain why/why not. If yes, draw the decision boundary.



2. With learning rate $\eta = 0.003$ and $W_{old} = 0.897$ derive and write new updated weight W_{new} using this loss function equation $L(x, \theta) = \left(\frac{1}{Y_{actual} - Y_{predicted}} \right)$.
3. Consider the below ROC curves for datasets with different numbers of negative examples. Which model would you prefer and why, considering the points on the curves with the same value of specificity 0.84 and sensitivity 0.5?



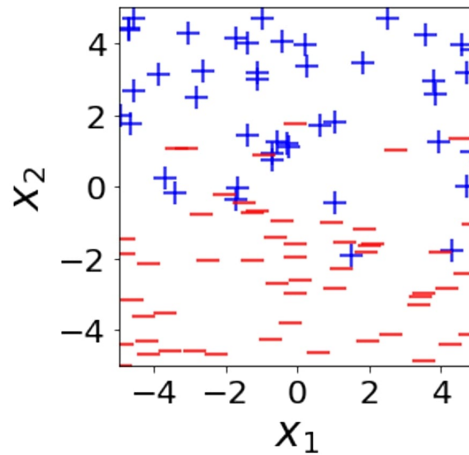
4. Write one example each of parametric and non-parametric algorithm.
5. You are analyzing data related to the sales of two products, Product A and Product B, over a period of 12 months. You have calculated various statistics and obtained the following results, justify your answer. 0 if not justified
- The average monthly sales of Product A is ₹10,000, and the standard deviation is ₹1,200.
 - The average monthly sales of Product B is ₹8,500, and the standard deviation is ₹900.
 - The covariance between the monthly sales of Product A and Product B is ₹15,000.

Select all the statements below that are true:

- (a) Product A and Product B have a positive relationship in sales because the covariance is positive.
 - (b) Product A consistently has higher sales than Product B because its average sales are higher.
 - (c) The standard deviation of Product A's sales is higher than the standard deviation of Product B's sales.
 - (d) The correlation between the monthly sales of Product A and Product B is -0.65.
6. For what value of K in K-means clustering algorithm, the error will be zero? Justify your answer in 1 line.
7. For the given data fit the linear regression model, find W_0 , W_1 and predict the missing value.

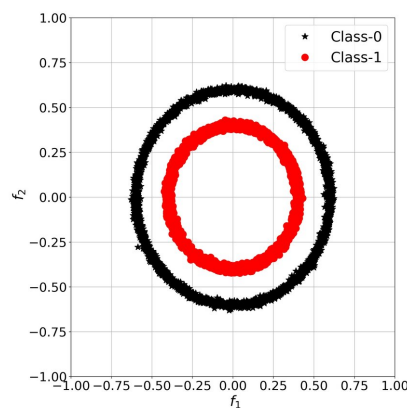
	1	2	3	4	5	6	7	8	9	10
X	1	3	4	6	7	8	9	12	13	15
Y	3.2	5.6	6.8	9.2	10.4	11.6	12.9	22	?	19

8. Which of the given below parameters best fit the given data distribution for logistic classifier, justify your answer. 0 if not justified



Tick the correct option.

- (a) $\theta_1 = 0, \theta_2 = 1, \theta_3 = 0$
 (b) $\theta_1 = -6, \theta_2 = 2, \theta_3 = 0$
 (c) $\theta_1 = 1, \theta_2 = 1, \theta_3 = 1$
 (d) $\theta_1 = -1, \theta_2 = 0, \theta_3 = 4$
9. Given a classification problem with input features as shown in the graph. Can we use logistic regression to classify this dataset? Explain why/why not. If yes, draw the decision boundary.



10. Shyam read about Generalized linear model (GLM) and made some assumptions, help shyam by commenting True/False on each statement. Justify your answers. **2 marks**
- (a) Target variable are independently distributed.
 (b) Target Variable must normally distributed
 (c) Non linear relationship between link function and explanatory variables
 (d) Errors must be normally distributed
11. Given the prediction data of a machine learning model, find (i) Precision (ii) Recall (iii) True Positive Rate (iv) F1 Score **2 marks**

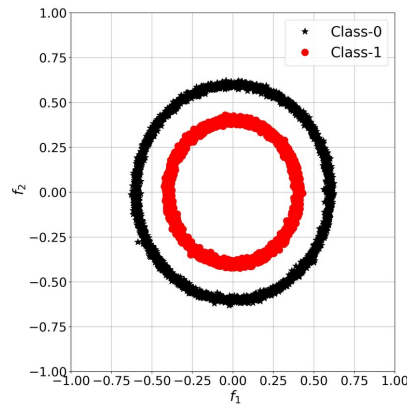
Predicted	Original
True	True
False	True
False	False
True	True
False	True
True	False
False	True

12. Shyam and Raju are working on some linear model, Shyam reported Pearson correlation coefficient value of +1.73 for its model, Raju is lazy and provided this data 2 marks

$(x - \bar{x})$	$(y - \bar{y})$
-3.2	-3.84
-1.2	-1.44
-0.2	-0.04
1.8	2.16
2.8	3.36

Calculate Pearson correlation coefficient for Raju and tell in which model x and y has strong positive linear relationship.

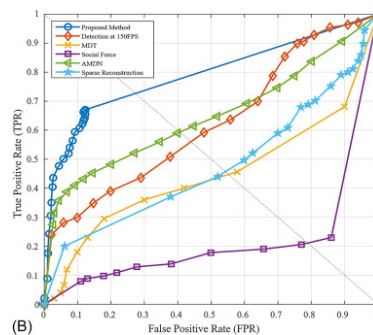
13. Derive weight update equation for linear regression using mean square error (MSE) loss function.
14. Plot an approximate transformation how the given data (Inner circle is class 1) is transformed in higher domain using gaussian kernel. Justify your answer. 2 marks



15. Derive OLS parameters for multiple linear regression.
16. Shyam selected some link functions to use in generalized linear model (GLM). help shyam by commenting on all functions whether to use or not. Justify your answers. 2 marks

- (a) $\log \frac{x_i}{1-x_i}$
- (b) $\log[\log[1 - x_i]]$
- (c) x^2
- (d) $\sin x$

17. Select which method has best Equal error rate. Justify your answer. 2 marks



18. A student has access to some data where features and targets are given, he applied principal component analysis (PCA) and claims he is using supervised learning. Is he correct, justify your answer?